



Fig. 15: Protective foil wrap of positive acting presensitized PCB type PS1015 from Kinsten (15cmx10cmx1.6mm)

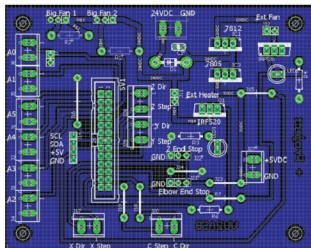


Fig. 16: Header pins aligned perfectly with the PCB holes

For ICs, see Fig. 13 for the default values used by Eagle and Fig. 14 for the recommended IC pad size and shape. The recommended pad diameter is 1.9304mm (0.076 inch) and pad drill size is 0.7mm (0.02755906 inch). Finally, check for electrical warnings and errors using the ERC and DRC capability of Eagle. Ensure that no airwires are left at the end of this step. The outcome should be similar to Fig. 16. The

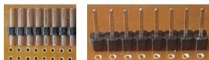


Fig. 17: Two views of header pins aligned perfectly with the PCB holes



Fig. 18: Kinsten board (left to right): White protective opaque sticker covering the presensitized side of the board, the PCB as seen from the component side and the presensitized surface revealed on peeling off the white sticker

next step depends on the technique you choose for developing the PCB. I have used various techniques and found the one using positive-acting pre-sensitized PCB laminates the best.

Its negative-acting pre-sensitized PCB variant requires a little different process for development. I chose the positive-acting board because it is easier to develop the artwork for it than for the negative-acting one. These boards can be bought online through many sources including eBay and Amazon.

Initially, I used

single-sided glass-epoxy positive-acting pre-sensitized board from Ever-Muse with excellent results (see Fig. 19). More recently, I used a similar board, Kinsten type PS1015, which cost me around ₹ 250 apiece. The board specifications are:

1. Dimensions: 10x15cm²
2. Thickness: 1.6mm
3. Material: Paper phenolic (PP), 1oz (35µm) copper foil, single side
4. Exposure time: 6-10 min-

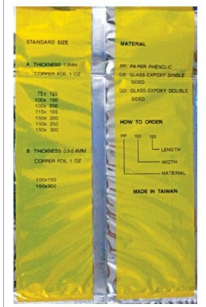


Fig. 19: Protective foil wrap of Ever-Muse glass-epoxy single-sided board (front and back)

utes using a 20W fluorescent lamp placed 5cm above the PCB or 60 to 90 seconds using UV exposure box

I applied the process described in this article on the Kinsten PS1015 positive-acting pre-sensitized PCBs. The process will probably work as such with almost any similar board. It may need to be tweaked for optimal results if the board specifications are quite different.

To be continued next month